A Universal Efficient Circular-Shift Network for Reconfigurable Quasi-Cyclic LDPC Decoders

Suwen Song, Hangxuan Cui, and Zhongfeng Wang

***Abstract*— Quasi-cyclic low-density parity-check (QC-LDPC) codes for modern communication standards usually have multiple code rates and block lengths. Therefore, reconfigurable LDPC decoders have received widespread attention, which require circular-shift networks to support various expansion factors. Besides, for inputs smaller than the network size, the circular-shift network is desired to process multiple frames in parallel to maximize hardware utilization efficiency. The increasing demands put severe challenges to low-complexity implementations of shift networks, especially for codes with numerous expansion factors, such as 5G LDPC codes. In this brief, we present a universal design of efficient reconfigurable circular-shift networks. Through an ingenious modification on the order of permutations, the generation of control signals is considerably simplified, leading to a significant reduction of area and critical path. Moreover, a hybrid architecture organically integrating different networks is proposed for further complexity reduction. Implementation results under TSMC 90 nm technology demonstrate that the proposed network can achieve 25% area reduction and 46% area-efficiency (AE) improvement over the state-of-the-art ones.**

***Index Terms*— 5G low-density parity-check (LDPC) codes, Banyan network (BN), barrel shifter, circular-shift networks, reconfigurable LDPC decoder.**

I. Introduction

Quasi-cyclic low-density parity-check (QC-LDPC) codes [1], owing to their remarkable error-correction performance and inherent parallelism for hardware implementation, have been widely adopted in modern wireless standards, such as IEEE802.11n [2], IEEE802.16e [3], and 5G [4]. In these communication standards, LDPC codes usually have multiple code rates and block lengths. For example, 5G LDPC codes have 51 expansion factors (Z) varying from 2 to 384 to pair with different code rates and lengths. Therefore, a reconfigurable LDPC decoder is desirable to support codes with different parameters. The circular-shift network is a fundamental module in LDPC decoders that performs circular-shift operations between check node units and variable node units according to shift values (SVs). For a reconfigurable LDPC decoder, the circular-shift network is required to support various expansion factors. Moreover, considering part of circuitry will stay idle if the expansion factor is less than the network size, the circular-shift network is preferably capable of processing multiple inputs with different SVs in parallel to maximize the hardware utilization efficiency.

An $N \times N$ Benes network [5] is composed of $2\log_2 N - 1$ stages of 2×2 switches and could perform any sort of permutations over N input data, where N should be a power of 2. In [6], the variation Benes network is proposed where 3×3 and 5×5 primitive switches are introduced to efficiently handle the case that the input size is not a power of 2. However, the variation Benes network [6] still suffers a long critical path and the generation of control signals is extremely sophisticated. Another alternative is the barrel shifter, in which the SVs are natural control signals thus facilitating controller design. By applying the bypass network to the barrel shifter, the QC-LDPC shift network (QSN) [7] could process input data of variable sizes, but is not able to process multiple frames in parallel. This issue is solved by the extended barrel-shifter (EBS) [8], but it requires a more complex generation of control signals than that for the QSN and the input frames must have the same SV. In [9], a fully parallel circular-shift rotation network is proposed. Based on a generic barrel shifter (GBS) core block, it could handle various frames with different permutation constraints. The main drawback of this design is that it is hard to perform run-time reconfiguration since the control signals need to be generated offline and stored in an extra read-only memory (ROM). The variation Banyan network (VBN) [10] is suitable for parallel processing due to its recursive structure and it is extended in [11] to support 5G standards. Lam *et al.* [12] also presents a segmented reconfigurable cyclic shifter for 5G LDPC decoders. However, when numerous operating modes need to be considered, their suboptimal controller designs still lead to a long critical path and large area consumption.

In this brief, we aim to develop a universal design of efficient circular-shift networks for reconfigurable LDPC decoders. Major contributions are summarized as follows.

1) Considering the complexity of the controller rises rapidly as the number of operating modes increases and thus deteriorates the throughput and area-efficiency (AE), we develop an improved Banyan network (IBN) by subtly adjusting the order of permutations, which facilitates a noticeable reduction on area consumption and critical path.
2) By analyzing the properties of different networks, a well-designed hybrid architecture is proposed, which is more flexible and efficient than prior works. Besides, it could provide a wide range of parallelism options covering all required expansion factors.
3) Following the proposed modifications, the reconfigurable circular-shift networks for IEEE802.11n and 5G standards are designed and implemented, which could provide 37% and 46% AE improvement over the existing works, respectively.

II. Background

The basic permutation networks could be classified into three categories: the Benes network, the Banyan network (BN), and the barrel shifter. Considering only the circular-shift operation is required in LDPC decoders while the Benes network is able to perform any kind of permutation, it is wasteful to apply the Benes network in LDPC decoders. Therefore, only the other two networks are briefly introduced in the following.

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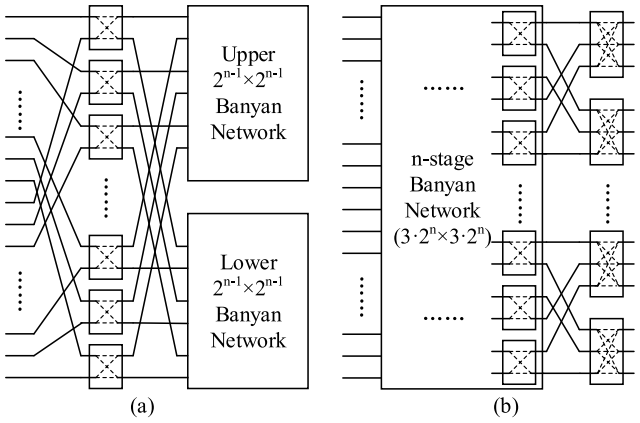


Fig. 1. (a) $2^n \times 2^n$ BN. (b) $3 \cdot 2^n \times 3 \cdot 2^n$ VBN.

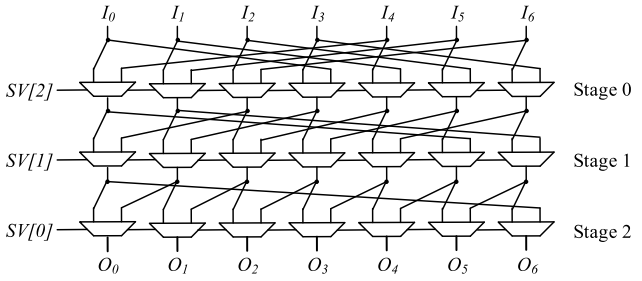


Fig. 2. 7×7 barrel shifter network.

A. Banyan Network

Fig. 1 shows the structures of the BN and the VBN. A $2^n \times 2^n$ BN is composed of n stages of 2×2 switch units. As shown in Fig. 1(a), the construction of the BN follows a recursive manner, so it is suitable for processing multiple frames in parallel. For example, if we set the former i stages of 2×2 switches to the bar state (do not switch), 2^i frames ($Z \leqslant 2^{n-i}$) could be processed in parallel. However, the BN is not flexible since the network size N could only be a power of 2. Consequently, part of components will be in idle state when the input size IN is not 2^n. The VBN is constructed by introducing a stage of 3×3 switch units to the conventional BN and it is more efficient to process data with $3 \cdot 2^n$ inputs, such as 5G LDPC codes ($Z_{\max} = 384 = 3 \times 2^7$). As shown in Fig. 1(b), a $3 \cdot 2^n \times 3 \cdot 2^n$ VBN consists of an n-stage BN which has $3 \cdot 2^n$ inputs and one stage of 3×3 switches.

B. Barrel Shifter Network

The predominant characteristic of the barrel shifter is that the SV is the natural control signal, and thus the hardware overhead of the control unit is negligible. Fig. 2 shows a 7×7 barrel shifter. As can be seen, SV[2], SV[1], and SV[0] are the control signals for each stage, which could circularly shift the input vector to left by 4, 2, and 1 times, respectively. Moreover, the network size of the barrel shifter could be an arbitrary value. However, the barrel shifter could only process one frame at a time, thus being inefficient when the input size is much smaller than the network size. EBS [8] could process several frames in parallel, but it requires a more complex generation of control signals and the input frames must have the same SV.

C. Bypass Network

When the input size IN is smaller than the network size N, the outputs of BN and barrel shifters would be ineffective due to the block problems [7], [10]. Therefore, the bypass network is proposed where two shifters are employed to eliminate the obstruction. One shifts the input data to right by SV times, while the other shifts the input data to left by IN–SV times. At last, a *Merge Network* extracts the useful part from the left-shift output and the right-shift output to comprise the final result. The applications of bypass network in the BN and barrel shifter could be found in [7]–[9].

Algorithm 1 Generation of Control Signals for the VBN

input : Shift Value SV, Network Size $IN = 3 \cdot 2^n$
initialize: $SV_1 = SV$
for $k = 1$ **to** $n - 1$ **do**
  if $SV_k < 3 \cdot 2^{n-k}$ **then**
    $Th_k = 3 \cdot 2^{n-k} - SV_k$
    $C_k[j] = 0$, for $0 \leqslant j < Th_k$
    $C_k[j] = 1$, for $Th_k \leqslant j < 3 \cdot 2^{n-k}$
  else
    $Th_k = 3 \cdot 2^{n-k+1} - SV_k$
    $C_k[j] = 1$, for $0 \leqslant j < Th_k$
    $C_k[j] = 0$, for $Th_k \leqslant j < 3 \cdot 2^{n-k}$
  $SV_{k+1} = SV\%(3 \cdot 2^{n-k})$
case SV_n
  0: $C_n = [0\ 1\ 0]$ 1: $C_n = [0\ 1\ 1]$ 2: $C_n = [0\ 0\ 1]$
  3: $C_n = [1\ 0\ 1]$ 4: $C_n = [1\ 0\ 0]$ 5: $C_n = [1\ 1\ 0]$
case $SV\%3$
  0: $C_{n+1} = 0$ 1: $C_{n+1} = 1$ 2: $C_{n+1} = 2$

III. Proposed Circular-Shift Network

For reconfigurable LDPC decoders, a well-designed circular-shift network should have the following characteristics: 1) could perform cyclic shift on input data with arbitrary size no larger than the network size (IN $\leqslant N$); 2) could process multiple input frames with different SVs in parallel; and 3) the complexity and critical path of the network are the smaller the better. To meet these purposes, we present an IBN and a hybrid circular-shift architecture in this section and realize the corresponding circular-shift networks for 5G and IEEE802.11n LDPC codes.

A. Improved BN

Let C_k denote the control signals of the kth stage in VBN, where $k = 1, 2, \ldots, n + 1$. Algorithm 1 gives the generation process of C_k when a single frame is processed in the VBN [10] with network size $3 \cdot 2^n$. Considering the control signals of the first stages of every subnetwork are the same, C_k is shared by 2^{k-1} Banyan subnetworks. According to Algorithm 1, C_k is determined by SV_k which equals $\mathrm{SV}\%(3 \cdot 2^{n+1-k})$. Therefore, n modular operations are introduced to the controller, resulting in a long critical path. When numerous operating modes need to be supported, the number of modular operations increases rapidly. As shown in Table I, 127 modular operations are needed for the permutation network presented in [11], which significantly aggravates the hardware overhead. Though a mapping scheme was proposed in [11] to eliminate modular operations, it requires lookup tables (LUTs) for data format conversions, thus inducing a larger area consumption.

The main reason for producing modular operations in VBN is that the size of the subnetwork is $3 \cdot 2^i$. Considering the modulo-2^i operation could be simply implemented by discarding the most significant bits (MSBs), through an ingenious modification on the

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Algorithm 2 Generation of Control Signals for the IBN

input : Shift Value SV, Network Size $IN = 3 \cdot 2^n$
1 **case** $SV[n+1:n]$
2 $\quad$ 0: $b = 1$ $\quad$ 1: $b = 2$ $\quad$ 2: $b = 0$
3 $Th_1 = 2^n - SV[n-1:0]$
4 $C_1[j] = SV[n+1:n]$, for $0 \leqslant j < Th_1$
5 $C_1[j] = b$, for $Th_1 \leqslant j < 2^n$
6 **for** $k = 2$ **to** n **do**
7 $\quad Th_k = 2^{n+1-k} - SV[n-k:0]$
8 $\quad C_k[j] = SV[n+1-k]$, for $0 \leqslant j < Th_k$
9 $\quad C_k[j] = \overline{SV[n+1-k]}$, for $Th_k \leqslant j < 2^{n+1-k}$
10 $C_{n+1} = SV[0]$

TABLE I
COMPLEXITY OF THE CONTROLLER IN CIRCULAR-SHIFT NETWORKS FOR THE RECONFIGURABLE 5G LDPC DECODER

	Modular	Sub.	Comp.	LUT3	LUT6
Proposed design	0	94	1536	1	0
Design in [11]	127	126	1599	128	64

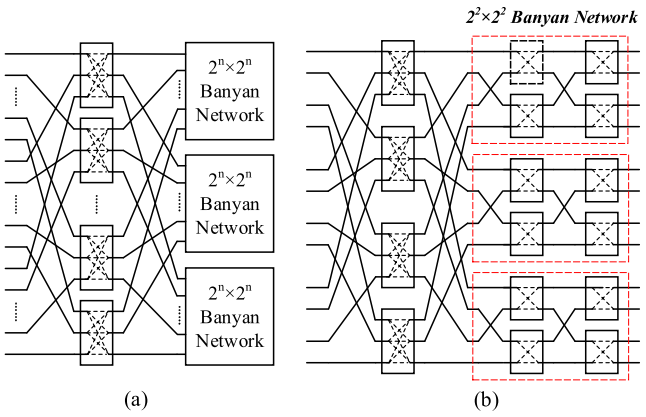


Fig. 3. (a) $3 \cdot 2^n \times 3 \cdot 2^n$ IBN. (b) 12×12 IBN.

order of permutations, we propose an IBN in which the sizes of all subnetworks are powers of 2, thus greatly reducing the critical path and computational complexity. Fig. 3(a) shows the structure of a $3 \cdot 2^n \times 3 \cdot 2^n$ IBN, where 2^n 3×3 switches are moved to the first stage followed by three $2^n \times 2^n$ BNs. It should be noted that the 3×3 switches could be replaced by 5×5 switches to support data with 5×2^n inputs.

Like the VBN, the proposed IBN could also be used for processing multiple frames with different SVs in parallel. For instance, Fig. 3(b) illustrates a 12×12 IBN, which could process three four-input frames simultaneously by setting the first stage of 3×3 switches to the bar state. The IBN has the same complexity as the VBN in datapath but owns a much simpler controller. Algorithm 2 details the generation of control signals for the $3 \cdot 2^n \times 3 \cdot 2^n$ IBN and it could be easily generalized to support other sizes of IBN. For a $3 \cdot 2^n \times 3 \cdot 2^n$ IBN, the SV should certainly be smaller than $3 \cdot 2^n$. Therefore, $SV[n+1:n]$ could only be 0, 1, and 2. Since the size of the subnetwork is 2^{n+2-k} $(k = 2, 3, \ldots, n+1)$, the control signals of the kth stage could be determined by $SV\%(2^{n+2-k})$ rather than by $SV\%(3 \cdot 2^{n+1-k})$ as in VBN. Considering $SV\%(2^{n+2-k}) = SV[n+1-k:0]$, all modular operations are eliminated. Moreover, from lines 2–9 of Algorithm 1 we can see that one multiplexer and

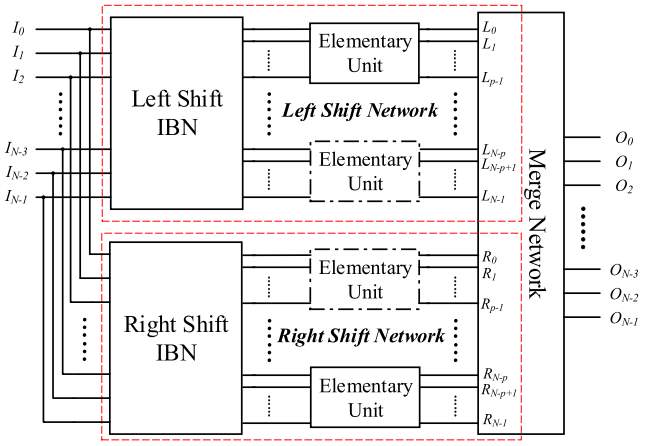


Fig. 4. Structure of the proposed hybrid circular-shift network.

two subtractions are used to determine Th_k, which is also shown in the following equation, where $N_s = 3 \times 2^{n-k+1}$ in VBN:

$$Th_k = \begin{cases} \dfrac{N_s}{2} - \mathrm{SV}_k, & \text{if } \mathrm{SV}_k < \dfrac{N_s}{2} \\ N_s - \mathrm{SV}_k, & \text{if } \mathrm{SV}_k \geqslant \dfrac{N_s}{2}. \end{cases} \tag{1}$$

When $\mathrm{SV}_k \geqslant (N_s/2)$, $N_s - \mathrm{SV}_k = (N_s - (N_s/2)) - (\mathrm{SV}_k - (N_s/2)) = (N_s/2) - (\mathrm{SV}_k - (N_s/2))$. Considering $N_s = 2^{n-k+2}$ and $\mathrm{SV}_k[n+1-k] = 1$ in IBN, $\mathrm{SV}_k - (N_s/2) = \mathrm{SV}_k[n-k:0]$, so $N_s - \mathrm{SV}_k = (N_s/2) - \mathrm{SV}_k[n-k:0]$. Moreover, since $\mathrm{SV}_k[n+1-k] = 0$ when $\mathrm{SV}_k < 2^{n-k+1}$, $(N_s/2) - \mathrm{SV}_k$ also equals $(N_s/2) - \mathrm{SV}_k[n-k:0]$. Therefore, one multiplexer and one subtraction could be saved in each stage, as shown in lines 3 and 7 of Algorithm 2.

The complexity reduction achieved by the proposed method is more significant for reconfigurable circular-shift networks since more control signals need to be generated. For the reconfigurable 5G LDPC decoder, we analyze the complexity of the controller in the IBN and compare it with the design presented in [11]. As shown in Table I, most of the modular operations, subtractions, three-input LUTs (LUT3s), and six-input LUTs (LUT6s) can be eliminated. The number of comparators is hard to be reduced since the majority of them (1536) are used to compare j with Th_k for determining $C_k[j]$, which is also needed in the proposed IBN.

B. Hybrid Circular-Shift Network Architecture

Based on the analysis in Section II, it could be concluded that the BN is suitable for parallel processing, the barrel shifter has the lowest complexity and the best flexibility, and the bypass network could be used for supporting variable input sizes. Therefore, to incorporate the advantages of different networks, we propose a hybrid circular-shift network which is more flexible and efficient than existing works.

Fig. 4 illustrates the structure of the proposed hybrid circular-shift network, which is constructed by recursively extending the elementary unit (EU) using IBNs. The IBN is used for parallel processing and the bypass network is also applied to handle the case when the input size is not exactly the network/subnetwork size. In the proposed network, EU is the minimum subnetwork, whose size is set as the smallest number among all required expansion factors, which is denoted by p. Since the EU will not be divided anymore, it is implemented by the barrel shifter for the lowest complexity. In summary, the proposed hybrid network could support arbitrary expansion factors included in the corresponding standards and provide a wide range of parallelism options while each input could possess a particular SV.

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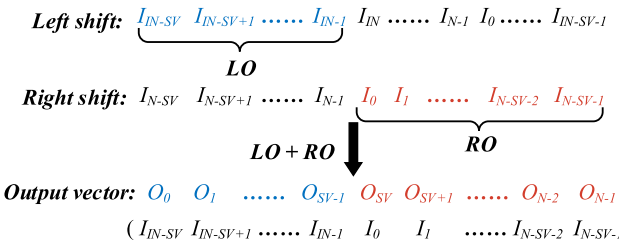


Fig. 5. Generation of the output vector when applying the bypass network.

The size of the proposed hybrid network could be $2^n \times p$, $3 \times 2^n \times p$, and $5 \times 2^n \times p$, so it is more flexible than the previous BNs, leading to a network size N closer to the input size IN. As stated in [6], an appropriate network size could reduce the number of idle components, thus alleviating the hardware overhead. For instance, in the case of the LDPC decoder for IEEE802.11n codes which have three expansion factors ($Z = 27, 54$, and 81), the sizes of circular-shift networks are 128, 96, and 81, respectively, when designed by BN, VBN, and the proposed hybrid network. Therefore, the number of multiplexers in the datapath could be reduced. It is worth mentioning that though the barrel shifter could also be used to minimize the idle circuitry, it could not perform parallel processing as the proposed one does.

Considering the SV is the natural control signal for EUs in the proposed network, only control signals for the IBN and *Merge network* need to be generated, which reduces the complexity and latency of the controller. A p-input EU could be implemented by $p \times b$ multiplexers, where $b = \lceil \log_2 p \rceil$. It has been proven in [7] that the number of multiplexers could be reduced to $p \times b - \sum_{i=0}^{b-1} 2^i$ when only the shift rather than the cyclic shift is required. However, since the EUs are also used to construct larger subnetworks, not all EUs could be simplified by only performing shifts. As shown in Fig. 5, the output vector of the network consists of a length-SV vector ***LO*** and a length-$(N - \mathrm{SV})$ vector ***RO***. When $(i-1) \cdot p \leqslant \mathrm{SV} < i \cdot p$ $(1 \leqslant i \leqslant N/p)$, ***LO*** includes all outputs of the former $i-1$ EUs and the first $\mathrm{SV} - (i-1) \cdot p$ outputs of the ith EU. Therefore, the former $i-1$ EUs should be capable of performing cyclic shift to form a larger frame. Since $i \leqslant N/p$, the 1st to the $(N/p - 1)$th EUs could not be simplified and only the last EU, which is labeled by dash-dotted line in the *Left Shift Network* of Fig. 4, could be simplified by only performing shift operation. Similarly, the first EU in the *Right Shift Network* could also be simplified, so $2 \times \sum_{i=0}^{b-1} 2^i$ multiplexers are saved in total.

C. Applications on 5G and IEEE802.11n LDPC Codes

In this subsection, we implement the proposed hybrid network for 5G and IEEE802.11n LDPC codes, as shown in Fig. 6. It should be noted that the proposed circular-shift network is not only effective for these two standards but is universal and could be generalized to other cases. For 5G LDPC codes, the maximum and minimum expansion factors are 384 and 2, respectively. Since p is very small ($p = 2$), EUs are integrated into IBNs so the corresponding hybrid network consists of two 3×2^7-input IBNs cascading a *Merge Network*. As shown in Table II, it could support all expansion factors in 5G standards and has eight parallelism options. At most 192 frames could be processed in parallel. For IEEE802.11n LDPC codes, the maximum and minimum expansion factors are 81 and 27, respectively. Therefore, the corresponding hybrid network consists of 3×2^0-input IBNs, 27-input EUs, and a *Merge Network*. As shown in Table II, it could process three size-27 frames simultaneously.

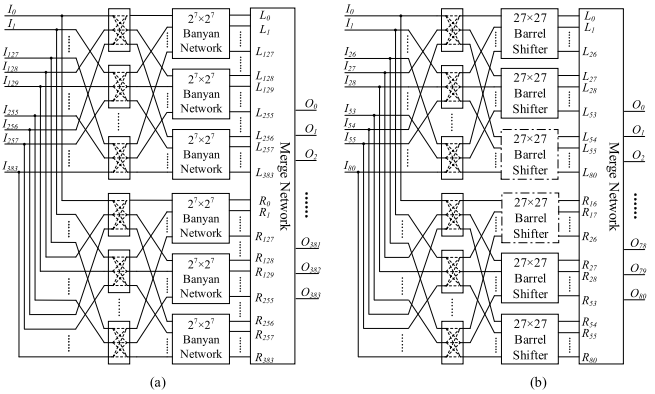


Fig. 6. (a) Proposed hybrid network for 5G LDPC decoders. (b) Proposed hybrid network for IEEE802.11n LDPC decoders.

TABLE II
PARALLELISM OPTIONS OF THE PROPOSED HYBRID NETWORKS

	Parallelism	Expansion Factors
5G	1	384 352 320 288 256 240 224 208 192 176 160 144
	3	128 120 112 104 96 88 80 72
	6	64 60 56 52 48 44 40 36
	12	32 30 28 26 24 22 20 18
	24	16 15 14 13 12 11 10 9
	48	8 7 6 5
	96	4 3
	192	2
IEEE802.11n	1	81 54
	3	27

TABLE III
ASIC SYNTHESIS RESULTS ON 90 nm CMOS TECHNOLOGY

	Proposed		[11]	Proposed		[10]
Standard	5G			IEEE802.11n		
Network Size	384×384			81×81		96×96
Message Bits	6-bit		6-bit	6-bit		6-bit
f (MHz)	775	552	552	800	600	600
Area (mm^2)	0.368	0.288	0.382	0.058	0.050	0.060
Gata Count ($\times 10^3$)	130.5	101.9	135.3	20.7	17.5	21.3
θ (Gbps)	298	212	212	64.8	48.6	48.6
AE (Gbps/mm^2)	809	736	555	1110	982	810

IV. IMPLEMENTATION RESULTS

The architectures presented in Section III-C are implemented and synthesized by Synopsys Design Compiler under the TSMC 90 nm 1.0 V technology. For fair comparisons, we also re-implement and synthesize the networks presented in [10] and [11] under the same environment, which are the state-of-the-art implementations of the circular-shift networks for reconfigurable 5G and IEEE802.11n LDPC decoders. All implemented networks could provide various parallelism options covering all expansion factors in the corresponding standard. Table III shows the comparison results, where f denotes the clock frequency, θ denotes the throughput, and AE denotes the area efficiency, which is calculated by θ/area. Considering the circular-shift network is an important module in LDPC decoders and lies in the critical path in most cases, it will limit the frequency and efficiency of the entire decoder if is not well designed. As shown in Table III, in comparison to [10], the network size of the proposed design could be smaller for the reconfigurable IEEE802.11n LDPC decoder. Moreover, though the proposed network has the same

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TABLE IV
AREA BREAKDOWN OF THE CIRCULAR-SHIFT NETWORKS

(mm^2)	Proposed		[11] or [10]	
	Datapath	Controller	Datapath	Controller
5G	0.243	0.045	0.303	0.079
IEEE802.11n	0.047	0.003	0.051	0.009

datapath delay as the VBN, since the critical path of the VBN is dominated by the controller and the proposed IBN and EU greatly simplify the generation of control signals, the frequency of the proposed design is improved by 33.3% and 40.4%, respectively, when compared to [10] and [11]. Considering the reduced area consumption, the AE of the proposed designs could achieve 1.37 and 1.46 times those in [10] and [11], respectively. Therefore, the proposed network could facilitate lower area consumption and higher clock frequency of the entire reconfigurable LDPC decoder.

When synthesized under the same frequency as those of the existing designs, the area of the proposed design could be smaller due to looser timing constraints. The comparisons in such case are also provided in Table III. As can be seen, the area consumptions of the proposed designs are 0.288 mm^2 and 0.050 mm^2 for 5G and IEEE802.11n LDPC decoders, which are 16.6% and 24.6% smaller than those of [10] and [11], respectively. Table IV shows the area breakdown, from which we can conclude that the complexity of controller is considerably reduced by applying the proposed optimizations. The area reduction in the datapath part mainly comes from the relatively small network size and the looser timing constraint which provides more flexibility to the Synopsys Design Compiler tool.

V. CONCLUSION

In this brief, we have proposed a universal efficient circular-shift network for reconfigurable LDPC decoders. Following an ingenious modification on the permutation order, the IBN is developed which significantly simplifies the generation of control signals. Moreover, to further improve the hardware efficiency, a hybrid architecture is presented by elaborately integrating barrel shifters and the proposed IBN. When applied in reconfigurable LDPC decoders, our approach could be used to improve the throughput by processing small codewords in parallel, thus facilitating base stations to support more low-traffic users. Experimental results demonstrate the advantages of the proposed design in benefiting lower area consumption and higher clock frequency of LDPC decoders.

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